# Scalability & Future Plan

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| **Date** | 29 June 2026 |
| **Team ID** | N/A |
| **Project Name** | Rising Waters – Flood Prediction |
| **MaXimum Marks** | 1 Mark |

## Scalability & Future Plan:

This document outlines how the **Rising Waters – Flood Prediction** system can be enhanced to support a larger number of users, process real-time environmental data, and integrate advanced disaster management technologies. It also presents the future roadmap for improving prediction accuracy, system performance, security, and accessibility.

## Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| 1 | No cloud deployment | Medium | High |
| 2 | No real-time weather API integration | High | High |
| 3 | Limited to single-user access | Medium | Medium |

## Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | Single User | Multi User support |

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|---|---|---|---|
| 2 | Data Storage | Local | Cloud Database (MySQL/Firebase/AWS) |
| 3 | Performance | Local processing | Cloud-based scalable processing and optimized ML inference |
| 4 | Security | Basic validation | User Authentication, Role-Based Access, and Data Encryption |

## Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | EXpected Impact |
|---|---|---|---|
| Phase 1 | Cloud Deployment (AWS, Azure, or Google Cloud) | 3 Months | Improves accessibility, enables remote access, and supports multiple users. |
| Phase 2 | Real-Time Weather API & River Sensor Integration | 6 Months | Provides live rainfall and river-level data, improving flood prediction accuracy. |
| Phase 3 | Mobile Application with Instant Flood Alerts | 9 Months | Delivers real-time flood notifications to users, disaster management teams, and local communities for faster emergency response. |